

ONEGA, Russian Federation

WMO: 226410

Lat: 63.909N

Lon: 38.123E

Elev: 42

StdP: 14.67

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-15.4	-26.6	1.3	-21.6	-20.0	1.8	-15.3	18.5	21.1	16.2	24.2	1.6	120	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	81.8	67.0	77.6	64.6	73.9	62.9	68.9	78.5	66.6	75.2	64.4	71.8	5.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	93.9	74.0	63.2	86.9	71.5	61.0	80.3	69.6	33.1	78.5	31.3	75.3	29.6	71.7	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.1	14.4	12.9	DB	-25.6	86.5	8.7	3.7	-31.9	89.1	-37.0	91.3	-41.8	93.4	-48.1	96.1	(4)
			WB	-25.8	72.1	8.4	3.1	-31.8	74.3	-36.7	76.1	-41.4	77.8	-47.5	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	12.5	13.5	22.7	34.6	46.0	56.2	62.8	57.9	49.0	38.1	26.3	19.7
	DBStd	19.62	14.17	13.99	9.36	8.62	9.63	8.45	7.46	6.43	6.14	7.06	9.77	12.67
	HDD50	5842	1161	1021	845	467	197	32	3	6	89	371	710	940
	HDD65	10413	1626	1441	1310	911	594	287	131	235	480	833	1160	1405
	CDD50	1012	0	0	0	6	72	218	400	253	60	3	0	0
	CDD65	107	0	0	0	0	4	23	63	17	0	0	0	0
	CDH74	789	0	0	0	1	39	199	437	111	2	0	0	0
	CDH80	171	0	0	0	0	6	43	100	22	0	0	0	0
Wind	WSAvg	5.3	5.0	5.1	5.6	5.1	5.8	5.3	4.9	4.6	5.1	5.8	5.7	5.8
Precipitation	PrecAvg	23.3	1.4	1.1	1.1	1.2	1.5	2.3	2.5	2.9	2.8	2.6	2.2	1.7
	PrecMax	29.8	2.6	2.4	2.6	3.0	2.9	5.4	4.8	5.9	5.5	4.9	3.9	3.4
	PrecMin	14.8	0.5	0.2	0.3	0.3	0.2	0.5	0.0	0.6	1.2	0.3	0.6	0.7
	PrecStd	3.0	0.6	0.6	0.6	0.7	0.6	1.3	1.1	1.3	1.1	0.9	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	38.5	46.0	67.0	79.5	84.9	87.3	83.2	70.7	57.0	46.5	41.0
		MCWB	34.9	36.0	38.5	51.2	60.7	68.9	69.5	68.4	61.0	52.8	44.1	39.0
	2%	DB	34.4	35.7	39.8	56.6	71.7	80.2	82.5	76.7	64.6	52.9	42.7	36.8
		MCWB	32.8	33.8	34.5	47.5	56.9	64.7	68.3	64.7	57.2	50.3	41.2	35.2
	5%	DB	32.7	33.6	37.3	50.4	66.9	75.1	79.0	72.2	61.2	50.4	40.0	35.5
		MCWB	31.6	32.0	33.6	43.1	54.6	61.7	66.5	62.7	55.7	48.2	38.7	34.2
	10%	DB	30.5	31.3	35.4	46.2	61.9	70.9	75.2	68.7	58.4	47.7	37.5	33.8
		MCWB	29.5	29.9	32.6	39.9	51.9	59.5	64.6	61.0	54.0	45.6	36.1	32.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	36.3	38.9	53.5	63.0	71.0	72.6	70.0	62.0	53.9	44.6	39.7
		MCDB	36.2	37.9	44.3	64.7	73.8	81.1	82.5	79.6	68.3	55.8	46.4	40.8
	2%	WB	33.0	34.0	35.7	47.9	59.0	66.8	69.9	66.5	58.5	50.9	41.2	35.6
		MCDB	34.1	35.5	38.4	56.4	68.1	77.7	79.6	73.9	63.0	52.7	42.6	36.4
	5%	WB	31.6	32.3	34.2	43.7	55.8	63.6	67.8	64.0	56.3	48.4	38.8	34.3
		MCDB	32.6	33.6	36.6	49.7	65.0	72.1	77.1	70.5	60.3	50.2	39.9	35.2
	10%	WB	29.5	29.8	32.8	40.5	52.8	60.4	65.9	61.9	54.3	45.7	36.3	32.7
		MCDB	30.6	31.1	35.2	45.3	61.1	69.2	73.7	67.7	58.1	47.4	37.3	33.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.6	13.5	13.6	15.1	15.3	14.8	13.6	11.3	7.2	7.2	8.6
		MCDBR	8.7	8.8	12.5	18.9	23.2	20.6	19.6	18.7	15.7	9.3	7.0	6.4
		MCWBR	8.3	8.0	9.3	11.7	12.4	10.3	9.6	9.6	9.7	7.4	6.8	6.0
	5% WB	MCDBR	8.4	8.8	10.3	18.1	20.7	17.6	17.2	16.1	13.5	8.8	7.1	6.3
		MCWBR	8.1	8.2	8.2	11.6	12.2	10.2	9.3	9.7	9.2	7.5	6.9	6.1
Clear-Sky Solar Irradiance	taub	0.232	0.288	0.316	0.353	0.357	0.351	0.387	0.364	0.336	0.311	0.369	N/A	
	taud	1.979	2.089	2.134	2.231	2.413	2.468	2.394	2.437	2.500	2.351	2.607	N/A	
	Ebn at Noon	104	192	240	259	271	276	260	256	240	190	100	N/A	
	Edh at Noon	9	21	30	35	32	30	32	28	22	16	8	N/A	
All-Sky Solar Radiation	RadAvg	48	218	661	1172	1502	1724	1609	1141	637	232	56	19	
	RadStd	5	43	81	100	146	161	175	127	73	33	11	2	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	+1.42	+1.76	N/A	N/A	N/A	N/A
Regional (5 neighbors)		+1.28	N/A	N/A	N/A	N/A	N/A	N/A	-374	N/A	N/A

Nomenclature: See separate page